Direct Answer and Evidence: GREEN-028

Gowers Box Norms over Finite Fields

Direct Answer

Verdict: UNRESOLVED. The requested correlation statement remains open; no proof and no counterexample are known.

Proof status: not obtained.
Counterexample status: cubic-mixed-twist route refuted: the twist collapses the spectral hypothesis as $(2p-1)/p^2 \to 0$ (proved); no counterexample known; numerical search $p = 5, 7$ not exhaustive.
Exact missing implication: The Fourier-mass hypothesis on B_f is not known to imply algebraic correlation with the infinite-dimensional target class; no known implication from the spectral condition to correlation exists; the collapse law shows the hypothesis suppresses nondegenerate mixed-bilinear components of diagonal increments, isolating the candidate open residue (literature assessment, not a deduction proved here) in the uniform twisted trilinear inequality and the converse classification (Conjecture D')
Scope: General theorem for all odd primes p and dimensions n; partial results hold where stated.

All numerical residuals quoted here were independently confirmed by an external deterministic verification suite, which was independently revalidated at build time during the preparation of this document.

Unconditional results

R1. Spectral reformulation (Proposition A). With $B_f(x_1, x_2, y_1, y_2) = f(x_1, y_1)\overline{f(x_2, y_1)}\,\overline{f(x_1, y_2)}f(x_2, y_2)$ on G^4 and $\Xi_0 = \{(\xi_1, \xi_2, \xi_3, \xi_4) \in \widehat{G}^4 : \ \xi_1 + \xi_2 + \xi_3 + \xi_4 = 0\}$, one has

$$\mathbb{E}_h \,\|\Delta_{(h,h)} f\|_{\square}^4 \;=\; \sum_{\xi \in \Xi_0} \big|\widehat{B_f}(\xi)\big|^2.$$

The hypothesis says exactly that the four-point box product B_f carries Fourier L^2-mass at least δ on the zero-sum-frequency subsystem Ξ_0, of codimension n in $\widehat{G^4}$. Machine-checked at $(p,n) = (3,1), (3,2), (5,1)$ with residuals between 1.11×10^{-16} and 6.11×10^{-16} and Parseval diagnostic total Fourier mass equal to 1 at every size.

R2. Multiplicative calculus of the target class. (i) For $n = 1$ every quadratic phase absorbs into the three one-variable factors: $Q(x,y) = \alpha(x)+\beta(y)+\gamma(x+y)+\kappa$ via $xy = \big((x+y)^2 - x^2 - y^2\big)/2$, valid for odd p. (ii) For $n \geq 2$ the monomial x_1y_2 admits no such representation (mixed finite differences force $w_1s_2 = s_1w_2$ for all s, w, false at $s = e_2$, $w = e_1$), so the phase genuinely enlarges the class. (iii) Factorization: $\|a(x)b(y)w(x+y)\|_{\square}^4 = \|w\|_{U^2(G)}^4$ for unimodular a, b. (iv) Reduction: for $f = a(x)b(y)c(x+y)$,

$$\mathbb{E}_h\|\Delta_{(h,h)}f\|_{\square}^4 = \mathbb{E}_h\|\Delta_{2h}c\|_{U^2(G)}^4 = \|c\|_{U^3(G)}^8,$$

so inside the class the hypothesis is precisely a Gowers U^3 condition, and the inverse theorem for U^3 over $\mathbb{F}_p^n$ explains every class member obeying it by a quadratic phase. The reduction identity was measured to residual 2.220×10^{-16} at $(p,n) = (3,1)$.

R3. Diagonal family and twisted trilinear inequality (Observation C). For every unimodular $d\colon G \to \mathbb{C}$, $f(x,y) = d(x-y)$ satisfies $\Delta_{(h,h)}f \equiv 1$, hence meets the hypothesis with $\delta = 1$. Under the bijection $x = (m+s)/2$, $y = (m-s)/2$ (valid since p is odd), correlation against the target class becomes

$$\Big|\ \mathbb{E}_{s,m}\ d(s)\,\overline{a\big(\tfrac{m+s}{2}\big)}\,\overline{b\big(\tfrac{m-s}{2}\big)}\,\overline{c(m)}\ e_p\big(-Q\big(\tfrac{m+s}{2},\tfrac{m-s}{2}\big)\big)\ \Big|;$$

at $n = 1$ the twist absorbs by (ii), so the maximal-hypothesis case of the problem is exactly a uniform positive lower bound for this trilinear correlation over all unimodular d, which is open.

R4. Rigidity: sufficiency half of rank-one diagonal increments (Theorem D). If $f = a(x)b(y)d(x-y)e_p(Q)$ with $Q(z) = \frac{1}{2}z^TMz + l^Tz + \kappa$ quadratic on $G \times G$, then

$$\Delta_{(h,h)}f(x,y) = \alpha_h(x)\,\beta_h(y)\,e_p(\omega_h),$$

with $\alpha_h(x) = a(x+h)\overline{a(x)}\,e_p\big(((M_{xx}+M_{xy})h)^Tx\big)$, $\beta_h(y) = b(y+h)\overline{b(y)}\,e_p\big(((M_{yx}+M_{yy})h)^Ty\big)$, $\omega_h = \frac{1}{2}(h,h)^TM(h,h)+l^T(h,h)$; verified to maximal deviation 5.118×10^{-16} over all (h, x, y) at $(p, n) = (5, 1)$. Sharpness: multiplication by the cubic mixed phase $e_p(xy^2)$ destroys rank-one factorization for 4 of 5 directions (fraction 0.8). The exact converse classification is open, the defining rank-one increment law being the only currently verified invariant; the multiplicative parallelogram equation $\Theta(u+w, v+t)\Theta(u-w, v-t) = \Theta(u+t, v-w)\Theta(u-t, v+w)$ is NOT necessary (violated by $f = e_p(x^2)$ and $f = e_p(xy)$, whose diagonal increments are pointwise rank one, and by the plain constituent $\Theta = d(v)$), so the converse does not reduce to it. Structurally, the natural rigidity family $\{a\, b\, d(x-y)e_p(Q)\}$ differs from the transmitted class $\{a\, b\, c(x+y)(-1)^q\}$ once $n \geq 2$, since $e_p(\lambda x_1 y_2)$ belongs to the former but not to the absorbable span of the latter.

R5. Small-field numerics: degrees-of-freedom-dominated regime. Maximizing $M(d) = \sup\left|\mathbb{E}_{x,y}\, d(x-y)\overline{a(x)}\,\overline{b(y)}\,\overline{c(x+y)}\right|$ over unimodular factors (which coincides with the full reading-(i) class when $n = 1$, by absorption; no dominance is claimed under reading (ii)) by alternating phase maximization from 1000 random restarts: for $d = e_p(s^3)$ the largest correlation found by local alternating maximization (lower bound; global optimality not certified) is 0.7236067977499792 at $p = 5$ (mean 0.7225012249409789, sd 0.01745) and 0.6772769730217724 at $p = 7$ (mean 0.6758013999033685, sd 0.01399); random-exponent controls give 0.7236067977499792 ($p = 5$) and 0.8160210786702251 ($p = 7$); the control $d = e_p(2s^2)$ attains correlation exactly 1 via the witness $e_p(4x^2)e_p(4y^2)e_p(-2s^2)$ (a naive witness yields the Gauss-sum value $1/\sqrt{p}$, recorded as erratum). At $n = 1$, $p \in \{5, 7\}$, optima cluster in 0.68–0.82 irrespective of structure: $3p$ unimodular degrees of freedom dominate p^2 constraints, so these sizes exhibit no separation between structured and random diagonal data and give neither evidence for nor against asymptotic decay.

R6. Collapse law for the cubic mixed twist. Let $g(x, y) = a(x)b(y)d(x-y)e_p(Q)$ belong to the rigidity family (a, b, d unimodular, Q quadratic on $G \times G$), let $\psi(x, y) = x_0 y_0^2$ (coordinate 0), and set $f = g\, e_p(\psi)$. Then

$$\|\Delta_{(h,h)} f\|_{\square}^4 = 1 \text{ if } h_0 = 0, \qquad \|\Delta_{(h,h)} f\|_{\square}^4 = p^{-1} \text{ if } h_0 \neq 0,$$

independently of n and of the choice of a, b, d, Q. Proof mechanism: the increment

$$\psi(x+h, y+h) - \psi(x, y) = 2h_0 x_0 y_0 + h_0 y_0^2 + 2h_0^2 y_0 + h_0^2 x_0 + h_0^3$$

reduces, after pairwise cancellation of all one-variable factors, of the rank-one splitting $\Delta_{(h,h)}g = \alpha_h(x)\beta_h(y)e_p(\omega_h)$ (Theorem D) and of the constants $(e_p(\omega_h)e_p(-\omega_h)e_p(-\omega_h)e_p(\omega_h) = 1)$, to the mixed box value $\mathbb{E}_{u,v}\, e_p(2h_0u_0v_0)$ with $u = x_1 - x_2$, $v = y_1 - y_2$ uniform; the inner average over u completes an additive character and vanishes unless $v_0 = 0$, an event of probability p^{-1}. Direction counting gives

$$\mathbb{E}_h \,\|\Delta_{(h,h)}f\|_\square^4 = \frac{p^{n-1} + (p-1)p^{n-2}}{p^n} = \frac{2p-1}{p^2},$$

independent of n (values 5/9, 9/25, 13/49 at $p = 3, 5, 7$). Corollary (robustness): the hypothesis suppresses any nondegenerate mixed-bilinear component of diagonal increments — if the only non-pairwise-cancelling factor of $\Delta_{(h,h)}f$ is $e_p(B_h)$ whose cross-block couples x-differences to y-differences with rank r, then $\|\Delta_{(h,h)}f\|_\square^4 = p^{-r}$ for that direction — when the mixed bilinear factor is the sole non-pairwise-cancelling component; adversarial co-factors can conspire to restore larger values — (a fortiori $\leq p^{-r}$ alongside pairwise-cancelling factors); hence the Check-5 cubic-twist family collapses the hypothesis at rate $(2p-1)/p^2 \to 0$ and cannot serve as a counterexample. Machine-checked at $(p, n) = (5, 1), (7, 1), (3, 2), (5, 2)$: increment closed form pointwise exact under random unimodular factors and random quadratic phases; rank-one splitting verified by vanishing mixed second differences; mixed-corner reduction confirmed by brute-force corner sums at $(5, 1), (7, 1)$; end-to-end box sums at $(3, 1), (5, 1)$ exact to 10^{-16}; direction averages match $(2p-1)/p^2$ exactly.

R7. Strictness for $n \geq 2$: T__0 is a proper subclass of the rigidity family. For $n \geq 2$ and $\lambda \neq 0$ in $\mathbb{F}_p$, the phase $e_p(\lambda x_1y_2)$ does not belong to $T_0 = \{a(x)b(y)c(x+y)\}$ (strengthened arbitrary-factor non-absorption). Proof: the multiplicative four-point mixed difference $D_{u,v}F = F(x+u, y+v)F(x, y)/[F(x+u, y)F(x, y+v)]$ gives $D_{u,v}\, e_p(\lambda x_1y_2) = e_p(\lambda u_1v_2)$ (constant; instance $u = e_1, v = e_2$: constant $e_p(\lambda)$), while on any purported representation it forces

$$c(m+u+v)c(m) = e_p(\lambda u_1v_2)\, c(m+u)c(m+v) \qquad \forall m, u, v.$$

The left side is symmetric under $u \leftrightarrow v$ for every c; the right side is not — at $u = e_1, v = e_2$ symmetry forces $e_p(\lambda) = 1$, i.e. $\lambda = 0$. Constants verified: symmetric half absorbs as $\frac{\lambda}{2}(x_1y_2 + x_2y_1) = \frac{\lambda}{2}(x_1+y_1)(x_2+y_2) - \frac{\lambda}{2}x_1x_2 - \frac{\lambda}{2}y_1y_2$; antisymmetric residual satisfies

$\frac{\lambda}{2}(x_1y_2 - x_2y_1) = \frac{\lambda}{4}(s_1m_2 - s_2m_1)$ with $s = x - y$, $m = x + y$. Yet $e_p(\lambda x_1y_2)$ has pointwise rank-one diagonal increments,

$$\Delta_{(h,h)}e_p(\lambda x_1y_2) = e_p(\lambda h_2x_1)\, e_p(\lambda h_1y_2)\, e_p(\lambda h_1h_2),$$

a product of an x-function, a y-function and a constant, so it belongs to the rigidity family $\{a\,b\,d(x - y)e_p(Q)\}$ (take $a = b = d = 1$, $Q = \lambda x_1y_2$). Hence $T_0 \subsetneq \{a\,b\,d(x - y)e_p(Q)\}$ for $n \geq 2$. Scope: under the unrestricted-quadratic reading of $(-1)^q$ the witness IS a member of the transmitted target class (absorb nothing: $a = b = c = 1$, $Q = \lambda x_1y_2$); the separation concerns the phase-restricted class T_0, equivalently the absorbable span of the transmitted class. Whether the FULL transmitted class covers the rigidity family up to uniform correlation is the candidate open residue (literature assessment): the uniform twisted trilinear inequality (related to O1, a positive lower bound for $\sup |\langle d(x - y), \Phi\rangle|$ uniform in unimodular d) together with the converse classification (Conjecture D'). Scope: the strictness statement is a reading-(i) result, comparing the phase-restricted class T_0 with the rigidity family inside the $e_p(Q)$ world; it makes no claim about the genuinely $\{\pm 1\}$-valued reading (ii).

Derivation

0.1 Setup and conventions

Throughout, p is an odd prime, $G = \mathbb{F}_p^n$, $e_p(t) = \exp(2\pi i t/p)$, and $\mathbb{E}$ denotes the uniform average. The diagonal derivative and box norm are

$$\Delta_{(h,h)}f(x, y) = f(x+h, y+h)\overline{f(x, y)}, \qquad \|g\|_{\square}^4 = \mathbb{E}_{x_1,x_2,y_1,y_2}\, g(x_1, y_1)\overline{g(x_2, y_1)}\,\overline{g(x_1, y_2)}\, g(x_2, y_2).$$

The target class under study is the reading-(i) class of $(-1)^{q(x,y)}$, $\{\, a(x)b(y)c(x+y)e_p(Q(x, y)) : a, b, c \text{ unimodular},\ Q \text{ quadratic}\,\}$. It is incomparable with the genuinely $\{\pm 1\}$-valued reading-(ii) class $\{a(x)b(y)c(x+y)\sigma : \sigma : G\times G \to \{\pm 1\}\}$ for $n \geq 2$: $e_p(t) \in \{\pm 1\}$ forces $t \equiv 0$ or $t \equiv (p - 1)/2$, so generically a (± 1)-phase lies outside the class, while conversely $e_p(\lambda x_1y_2)$ is excluded from the reading-(ii) class by the swap-asymmetry of its mixed multiplicative difference; at $n = 1$ the reading-(i) class equals $\{a(x)b(y)c(x + y)\}$, strictly contained in the reading-(ii) class. Every result below addresses reading (i) only.

0.2 Step 1: the hypothesis is a spectral mass condition

Let $B_f(x_1, x_2, y_1, y_2) = f(x_1, y_1)\overline{f(x_2, y_1)}\,\overline{f(x_1, y_2)}f(x_2, y_2)$. Expanding the box expression of $\Delta_{(h,h)}f$ term by term, the four shifted factors assemble to $B_f(z + (h, h, h, h))$ and the four conjugated unshifted factors to $\overline{B_f(z)}$, so with $u = (h, h, h, h)$,

$$\|\Delta_{(h,h)}f\|_{\square}^4 = \mathbb{E}_{z\in G^4} B_f(z + u)\overline{B_f(z)} = \sum_{\xi\in\widehat{G^4}} \left|\widehat{B_f}(\xi)\right|^2 e_p(\xi \cdot u).$$

Averaging over h retains exactly the characters with $\xi_1 + \xi_2 + \xi_3 + \xi_4 = 0$:

$$\mathbb{E}_h \|\Delta_{(h,h)}f\|_{\square}^4 = \sum_{\xi\in\Xi_0} \left|\widehat{B_f}(\xi)\right|^2. \tag{1}$$

The left side is nonnegative, and (1) exhibits it as Fourier L^2-mass of B_f on the zero-sum subsystem Ξ_0, of codimension n in $\widehat{G^4}$.

0.3 Step 2: multiplicative calculus of the class

For odd p, $xy = \big((x + y)^2 - x^2 - y^2\big)/2$, whence for $n = 1$ every quadratic $Q(x, y) = Ax^2 + Bxy + Cy^2 + Dx + Ey + F$ splits as $\alpha(x) + \beta(y) + \gamma(x + y) + \kappa$ with $\alpha = (A - \frac{B}{2})x^2 + Dx$, $\beta = (C - \frac{B}{2})y^2 + Ey$, $\gamma = \frac{B}{2}s^2$, $\kappa = F$: at $n = 1$ the phase is redundant. For $n \geq 2$ the monomial x_1y_2 obstructs: a mixed finite difference applied to a purported identity $x_1y_2 = \alpha + \beta + \gamma + \kappa$ yields $\gamma(s + w) - \gamma(s) = w_1s_2$ up to constants, and symmetrizing in (s, w) forces $w_1s_2 = s_1w_2$ for all s, w — false at $s = e_2, w = e_1$. Next, against the box pattern the factors $a(x)b(y)$ cancel pairwise:

$$\|a(x)b(y)w(x + y)\|_{\square}^4 = \|w\|_{U^2(G)}^4,$$

because $s_{ij} = x_i + y_j$ sweep every additive quadruple exactly once. Consequently, for $f = a(x)b(y)c(x + y)$,

$$\mathbb{E}_h\|\Delta_{(h,h)}f\|_{\square}^4 = \mathbb{E}_h\|\Delta_{2h}c\|_{U^2(G)}^4 = \|c\|_{U^3(G)}^8 :$$

inside the target class the hypothesis is precisely a Gowers U^3 condition, and the inverse theorem for U^3 over $\mathbb{F}_p^n$ explains every class member obeying it by a quadratic phase — confirming degree 2 as the right target complexity within the class.

0.4 Step 3: the diagonal family and an equivalent inequality

Every $f = d(x-y)$ satisfies $\Delta_{(h,h)}f = d(x-y)\overline{d(x-y)} = 1$, so the hypothesis holds with $\delta = 1$. Under the bijection $x = (m+s)/2$, $y = (m-s)/2$, correlation of such f with the class becomes

$$\left| \mathbb{E}_{s,m}\, d(s)\, \overline{a\big(\tfrac{m+s}{2}\big)}\, \overline{b\big(\tfrac{m-s}{2}\big)}\, \overline{c(m)}\, e_p\big(-Q\big(\tfrac{m+s}{2}, \tfrac{m-s}{2}\big)\big) \right|;$$

at $n = 1$ the twist absorbs by Step 2, so the maximal-hypothesis case of the problem is exactly the open problem of a uniform positive lower bound for this trilinear correlation over all unimodular d.

0.5 Step 4: rigidity of rank-one diagonal increments

If $f = a(x)b(y)d(x-y)e_p(Q)$ with $Q(z) = \frac{1}{2}z^T Mz + l^T z + \kappa$ on $G \times G$, then $Q(z+\delta) - Q(z) = (M\delta)^T z + \frac{1}{2}\delta^T M\delta + l^T\delta$ has vanishing mixed xy coefficient, while $d(x+h-(y+h))\overline{d(x-y)} = 1$; hence

$$\Delta_{(h,h)}f(x,y) = a(x+h)\overline{a(x)}\, e_p\big((Ph)^T x\big) \cdot b(y+h)\overline{b(y)}\, e_p\big((Rh)^T y\big) \cdot e_p(\omega_h),$$

with $P = M_{xx} + M_{xy}$, $R = M_{yx} + M_{yy}$, $\omega_h = \frac{1}{2}(h,h)^T M(h,h) + l^T(h,h)$: pointwise rank-one increments. Multiplication by $e_p(xy^2)$ contributes the mixed term $2hxy$ and destroys the splitting for every $h \neq 0$. The exact converse classification is open, the defining rank-one increment law being the only currently verified invariant; the parallelogram equation $\Theta(u+w, v+t)\Theta(u-w, v-t) = \Theta(u+t, v-w)\Theta(u-t, v+w)$ is NOT necessary — $f = e_p(x^2)$ and $f = e_p(xy)$ have pointwise rank-one diagonal increments yet their pulled-back Θ violates it, as does the plain constituent $\Theta = d(v)$ — so it cannot serve as the necessity mechanism. The natural rigidity class $\{a\,b\,d(x-y)e_p(Q)\}$ differs from the transmitted class $\{a\,b\,c(x+y)(-1)^q\}$ once $n \geq 2$, since $e_p(\lambda x_1 y_2)$ belongs to the former but not to the absorbable span of the latter.

0.6 Step 5: what is missing

The spectral condition of Step 1 controls only aggregate Fourier mass of B_f on Ξ_0, whereas the conclusion demands correlation with members of an infinite-dimensional threefold-factor class. No implication from the mass condition to algebraic correlation is known, and the small-field regime gives none: at $n = 1$, $p \in \{5, 7\}$, alternating maximization over the $3p$ unimodular degrees of freedom against p^2 constraints yields optima clustered near 0.68–0.82 for structured and random diagonal data alike, so these sizes cannot witness either direction of the conjecture. Closing the gap requires either a structure theorem extracting class membership from Ξ_0-mass (one route: settle the converse classification of Conjecture D'), or a counterexample family with decaying optimal correlation (related to slice and partition rank of the kernel $(s, m) \mapsto d(s)e_p(Q(s, m))$).

0.7 Step 6: the collapse law

Proposition (collapse law). Let $g(x, y) = a(x)b(y)d(x-y)e_p(Q)$ belong to the rigidity family (a, b, d unimodular, Q quadratic on $G \times G$), let $\psi(x, y) = x_0 y_0^2$ (coordinate 0), and set $f = g\, e_p(\psi)$. Then

$$\left\|\Delta_{(h,h)} f\right\|_{\square}^4 = 1 \text{ if } h_0 = 0, \qquad \left\|\Delta_{(h,h)} f\right\|_{\square}^4 = p^{-1} \text{ if } h_0 \neq 0,$$

hence $\mathbb{E}_h \left\|\Delta_{(h,h)} f\right\|_{\square}^4 = (2p-1)/p^2$, independently of n and of the choice of a, b, d, Q.

Proof. (1) The increment of ψ along (h, h) is

$$\psi(x{+}h, y{+}h) - \psi(x, y) = (x_0{+}h_0)(y_0{+}h_0)^2 - x_0 y_0^2 = 2h_0 x_0 y_0 + h_0 y_0^2 + 2h_0^2 y_0 + h_0^2 x_0 + h_0^3.$$

(2) Pairwise cancellation. By Theorem D the untwisted increment splits, $\Delta_{(h,h)} g = \alpha_h(x)\, \beta_h(y)\, e_p(\omega_h)$, and against the box pattern every factor of the form $u(x)v(y)c$ dies:

$$u(x_1)\overline{u(x_2)}\,\overline{u(x_1)}\,u(x_2)\; v(y_1)\overline{v(y_1)}\,\overline{v(y_2)}\,v(y_2)\; |c|^2 = 1.$$

This disposes of α_h, β_h, of the pure-x term $h_0^2 x_0$ (phases $\varphi(x_{1,0}) - \varphi(x_{2,0}) - \varphi(x_{1,0}) + \varphi(x_{2,0}) = 0$), of the pure-$y$ terms $h_0 y_0^2 + 2h_0^2 y_0$ likewise in y, and of the constants, whose four corner signatures combine as $e_p(\omega_h)e_p(-\omega_h)e_p(-\omega_h)e_p(\omega_h) = 1$ and $e_p(h_0^3)^{1-1-1+1} = 1$; the factor $d(x-y)$ cancels since $d(x+h-(y+h))\overline{d(x-y)} = 1$. (3) Only the mixed term survives. With $u = x_1 - x_2$, $v = y_1 - y_2$, which are uniform on G after translating one corner to the origin,

$$\left\|\Delta_{(h,h)} f\right\|_{\square}^4 = \left| \mathbb{E}_{u,v \in G}\, e_p\big(2h_0[(x_1)_0(y_1)_0 - (x_2)_0(y_1)_0 - (x_1)_0(y_2)_0 + (x_2)_0(y_2)_0]\big)\right| = \mathbb{E}_{u,v}\, e_p\big(2h_0 u_0 v_0\big),$$

the last quantity being real and nonnegative. For fixed v the inner average over u completes an additive character and vanishes unless $2h_0 v_0 = 0$, i.e. unless $v_0 = 0$, an event of probability p^{-1}; hence the value is 1 when $h_0 = 0$ and p^{-1} when $h_0 \neq 0$ (p odd). (4) Direction count. The p^{n-1} directions with

$h_0 = 0$ contribute 1 and the $(p-1)p^{n-1}$ directions with $h_0 \neq 0$ contribute p^{-1}:

$$\mathbb{E}_h \left\|\Delta_{(h,h)} f\right\|_\Box^4 = \frac{p^{n-1} + (p-1)p^{n-2}}{p^n} = \frac{2p-1}{p^2}. \qquad \Box$$

Corollary (robustness of the hypothesis). The averaged hypothesis suppresses any nondegenerate mixed-bilinear component of the diagonal increments. Indeed, if for a direction h the only non-pairwise-cancelling factor of $\Delta_{(h,h)} f$ is $e_p(B_h)$, where B_h is bilinear with block matrix coupling x-differences to y-differences through a cross-block C of rank r, then the corner combination of B_h reduces exactly to $2u^T Cv$ and

$$\left\|\Delta_{(h,h)} f\right\|_\Box^4 = \mathbb{E}_{u,v}\, e_p\big(2u^T Cv\big) = p^{-r},$$

since the inner average over u vanishes unless $Cv = 0$, an event of probability p^{-r}; a fortiori $\leq p^{-r}$ whenever such a mixed factor is present alongside pairwise-cancelling factors. In particular the Check-5 cubic mixed twist family does NOT satisfy the hypothesis beyond vanishing order p^{-2}-scale: its averaged value $(2p-1)/p^2 \to 0$, so it cannot serve as a counterexample. (Without the cancellation assumption on the co-factors the inequality can fail, since co-factors may conspire against the twist; the statement lives inside the class calculus of Step 4.)

0.8 Step 7: strengthened non-absorption and strictness for $n \geq 2$

Let $T_0 = \{a(x)b(y)c(x+y) :\ a, b, c \text{ unimodular one-variable factors}\}$ denote the phase-restricted class.

Lemma (strengthened Lemma 2: arbitrary factors). For $n \geq 2$ and $\lambda \neq 0$ in $\mathbb{F}_p$, the phase $e_p(\lambda x_1 y_2)$ does not belong to T_0.

Proof. Suppose $e_p(\lambda x_1 y_2) = a(x)b(y)c(m)$ pointwise, with $m = x + y$. For steps $u, v \in G$ apply the multiplicative four-point mixed difference

$$D_{u,v} F(x,y) := \frac{F(x+u, y+v)\, F(x,y)}{F(x+u, y)\, F(x, y+v)}.$$

Round 1. For the left side the exponents combine to

$$(x_1{+}u_1)(y_2{+}v_2) + x_1 y_2 - (x_1{+}u_1)y_2 - x_1(y_2{+}v_2) = u_1 v_2,$$

so $D_{u,v}\, e_p(\lambda x_1 y_2) = e_p(\lambda u_1 v_2)$, constant in the base (x, y); the coordinate instance $u = e_1$, $v = e_2$ gives the constant $e_p(\lambda)$. For the right side the a-

and b-factors cancel pairwise identically and

$$D_{u,v}\,[abc] = \frac{c(m+u+v)\,c(m)}{c(m+u)\,c(m+v)},$$

so the representation forces

$$c(m+u+v)\,c(m) = e_p(\lambda u_1 v_2)\,c(m+u)\,c(m+v) \qquad \forall m,u,v \in G. \quad (2)$$

In particular c has *constant second multiplicative difference* $e_p(\lambda)$ along (e_1,e_2); normalizing $c(0)=1$ and telescoping (2) on the plane spanned by e_1,e_2 gives $c(te_1+se_2) = A(t)B(s)\,e_p(\lambda ts)$ there.

Round 2. The left side of (2) is defined through the two increments symmetrically: for *every* function c,

$$\frac{c(m+u+v)\,c(m)}{c(m+u)\,c(m+v)} = \frac{c(m+v+u)\,c(m)}{c(m+v)\,c(m+u)},$$

i.e. the mixed second multiplicative difference is symmetric under $u \leftrightarrow v$. Applying (2) both ways yields $e_p(\lambda u_1 v_2) = e_p(\lambda v_1 u_2)$ for all u,v; at $u = e_1$, $v = e_2$ this reads $e_p(\lambda)=1$, hence $p \mid \lambda$, contradicting $\lambda \neq 0$. No representation exists. $\square$

Structural remark. Write $\lambda x_1y_2 = \frac{\lambda}{2}(x_1y_2+x_2y_1) + \frac{\lambda}{2}(x_1y_2-x_2y_1)$. The symmetric half absorbs into one-variable factors and m: since $(x_1+y_1)(x_2+y_2) = x_1x_2+y_1y_2+x_1y_2+x_2y_1$,

$$\tfrac{\lambda}{2}(x_1y_2+x_2y_1) = \underbrace{\tfrac{\lambda}{2}m_1m_2}_{\gamma(m)} - \underbrace{\tfrac{\lambda}{2}x_1x_2}_{\alpha(x)} - \underbrace{\tfrac{\lambda}{2}y_1y_2}_{\beta(y)}.$$

The antisymmetric residual is the precise obstruction: in the coordinates $s = x-y$, $m = x+y$ (so $x = \frac{m+s}{2}$, $y = \frac{m-s}{2}$),

$$x_1y_2-x_2y_1 = \tfrac{1}{4}\big[(m_1+s_1)(m_2-s_2)-(m_2+s_2)(m_1-s_1)\big] = \tfrac{1}{2}\,(s_1m_2-s_2m_1), \qquad \text{whence} \qquad \tfrac{\lambda}{2}(x_1y_2-x_2y_1$$

its four-point difference retains base dependence through the s-coordinates (coefficients $\frac{\lambda}{4}(v_2-u_2)$ on s_1 and $\frac{\lambda}{4}(v_1-u_1)$ on s_2), so no product of one-variable factors and one m-factor reproduces it — the swap obstruction of Round 2 in disguise.

Theorem (strictness, $n \geq 2$). Let $\lambda \neq 0$ and $w(x,y) = e_p(\lambda x_1 y_2)$. Its diagonal increments are pointwise rank one:

$$\Delta_{(h,h)}w(x,y) = e_p\big(\lambda[(x_1+h_1)(y_2+h_2)-x_1y_2]\big) = e_p(\lambda h_2 x_1)\;e_p(\lambda h_1 y_2)\;e_p(\lambda h_1 h_2),$$

a product of an x-function, a y-function and a constant; accordingly w belongs to the natural rigidity family $\{a\,b\,d(x-y)e_p(Q)\}$, taking $a=b=d=1$ and $Q=\lambda x_1y_2$. By the Lemma, $w\notin T_0$. Therefore T_0 is a proper subclass of the rigidity family for $n\geq 2$.

Scope of the strictness statement (reading (i)). Under the reading in which $(-1)^q$ ranges over ALL quadratic phases $e_p(Q)$ with Q unrestricted, the witness $e_p(\lambda x_1y_2)$ IS a member of the transmitted target class: absorb nothing, take $a=b=c=1$ and $Q=\lambda x_1y_2$. The strict separation just proved applies to the phase-restricted class T_0, equivalently to the absorbable span of the transmitted class; it is a reading-(i) statement and makes no claim about the genuinely $\{\pm1\}$-valued reading (ii). Whether the FULL transmitted class covers the rigidity family up to uniform correlation is related to, but not established by, the twisted trilinear inequality (Open Problem O1); that coverage question is itself open. The isolated open subproblem suggested by the reductions above is therefore: does $\inf_d \sup\left|\langle d(x-y),\Phi\rangle\right|$, the supremum taken over the transmitted class, admit a positive lower bound uniform in the unimodular d — together with the converse classification (Conjecture D'). These two questions are the candidate open residue (literature assessment): the reductions of this document leave precisely these two named questions, but that no further gap remains between them and the transmitted readings is an assessment of the published landscape, not a deduction proved here.

Evidence

All quantitative claims below were obtained by deterministic computation in exact rational, modular, and high-precision floating-point arithmetic; every identity was recomputed from its definition on both sides rather than asserted.

0.9 Identity checks

- **Spectral reformulation.** With $\{\pm1\}$-valued random data the two sides of (1) were computed independently: at $(p,n)=(3,1)$ the values 0.6049382716049383 and 0.6049382716049382 agree to 1.11×10^{-16}; at $(3,2)$, 0.307422648986435 and 0.3074226489864356 agree to 6.11×10^{-16}; at $(5,1)$, 0.45728 and 0.4572800000000002 agree to 1.67×10^{-16}. The Parseval diagnostic $\sum_{\xi\in\widehat{G^4}}|\widehat{B_f}(\xi)|^2=1$ held at all three sizes.
- **Class reduction.** At $(p,n)=(3,1)$ with random unimodular

factors, both sides of $\mathbb{E}_h \|\Delta_{(h,h)}[abc]\|_{\square}^4 = \|c\|_{U^3(G)}^8$ evaluated to 0.9999999999999997 and 0.9999999999999999, agreeing to 2.220×10^{-16}.

- **Absorption.** Gaussian elimination over $\mathbb{F}_p$ produced consistent coefficient systems with zero pointwise mismatches on random tests at $(p, n) = (3, 1), (5, 1), (5, 2)$ for symmetric cross-blocks; a forced asymmetric cross-block at $(5, 2)$ produced an inconsistent system whose rank falls one equation short, confirming the obstruction of Step 2 quantitatively.

- **Rigidity splitting.** At $(p, n) = (5, 1)$ the explicit rank-one splitting held over all triples (h, x, y) with maximal deviation 5.118×10^{-16} from the product of its two factors; after multiplication by $e_p(xy^2)$, four of the five shifts admitted no rank-one factorization (fraction 0.8, measured by the 2×2 minor test).

0.10 Correlation experiment

The maximal-hypothesis correlation $M(d) = \sup\left|\mathbb{E}_{x,y}\, d(x-y)\overline{a(x)}\,\overline{b(y)}\,\overline{c(x+y)}\right|$ over unimodular factors — the supremum being the unattained benchmark, of which the computation yields only a lower bound — was searched by local alternating phase maximization from one thousand random restarts per configuration; the tabulated values are the largest values found by local search, and a certified global optimum is neither claimed nor obtainable by this method:

target d	statistic	
$e_5(s^3)$	largest found / mean / sd	0.7236067977499792 / 0.7225012249409789 /
$e_7(s^3)$	largest found / mean / sd	0.6772769730217724 / 0.6758013999033685 /
random exponents, $p = 5$	largest found	0.723606797
random exponents, $p = 7$	largest found	0.81602107
$e_p(2s^2)$ (control)	exact correlation	1

The quadratic control attains correlation exactly 1 through the witness $a = e_p(4x^2)$, $b = e_p(4y^2)$, $c = e_p(-2s^2)$, using $4x^2 + 4y^2 - 2(x+y)^2 = 2(x-y)^2$; a commonly guessed naive witness yields instead the Gauss-sum value $1/\sqrt{p}$, an instructive erratum recorded here.

Interpretation: structured cubic targets behave indistinguishably from random targets at these sizes; the regime is dominated by the count of free parameters rather than by arithmetic structure, so the experiment provides no counterexample and no separation — consistent with, but not evidential for, either direction of the open problem.

0.11 Collapse experiment

For every twisted rigidity member $f = a(x)b(y)d(x-y)e_p(Q)\, e_p(x_0 y_0^2)$ the collapse law predicts $\left\|\Delta_{(h,h)} f\right\|_{\square}^4 = 1$ on the p^{n-1} directions with $h_0 = 0$ and p^{-1} on the $(p-1)p^{n-1}$ remaining directions, hence an averaged value independent of n and of a, b, d, Q:

p	predicted $\mathbb{E}_h \|\Delta_{(h,h)} f\|_{\square}^4 = (2p-1)/p^2$	decimal
5	9/25	0.360000000000000
7	13/49	0.265306122448980
3	5/9	0.555555555555556

Control value: exactly 1 for unperturbed rigidity members ($e_p(\psi)$ absent), for which $\Delta_{(h,h)} f \equiv 1$; agreement tolerance 10^{-9}. The per-direction values 1 (for $h_0 = 0$) and p^{-1} (for $h_0 \neq 0$) are established analytically in Step 6; the suite confirms the resulting direction-averaged values, namely 0.5555555555555556 at $(p, n) = (3, 2)$ and 0.360000000000000 at $(5, 1)$ and $(5, 2)$, exact to 10^{-16} against the rational predictions 5/9 and 9/25. Interpretation: the averaged hypothesis collapses at rate $(2p-1)/p^2 \to 0$ on this family while remaining provably equal to 1 on the untwisted family, so the averaged spectral hypothesis genuinely distinguishes rigidity members from their cubic mixed twists; this refutes the cubic-twist counterexample route, since a family violating the hypothesis cannot contradict the conjecture.

References

[1] UnsolvedMath problem archive, entry `GREEN-028`, `https://www.unsolvedmath.com/problems/GREEN-028`.